Sharva Gogawale

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EDUCATION

Tel Aviv University — **Tel Aviv, Israel**

Master of Science in Electrical and Computer Engineering (GPA: 92.9/100) — *April 2026*

Selected Coursework: Deep Learning, Computer Vision, Deep Learning for Medical Imaging, Optimization, Optimal Control

Bachelor of Science in Electrical and Electronics Engineering — *July 2022*

Selected Coursework: Data Structures & Algorithms, Digital Signal Processing, Random Signals & Noise

SELECTED PUBLICATIONS

[J: Journal, C: Conference]

[C8] *The Usual Positives: Selecting Supervision for Figurative Language Detection.* **Sharva Gogawale**, Amutheezan Sivagnanam, Nachum Dershowitz, Rakesh M Verma. ACL ARR October 2026 Submission.

[C7] *Beyond Literal Meaning: Figure-of-Speech-Aware Deception Detection.* Amutheezan Sivagnanam, **Sharva Gogawale**, Rakesh M Verma. ACL ARR August 2026 Submission.

[C6] *Auditing Gendered Name Sensitivity in LLM-Based Candidate Screening.* **Sharva Gogawale**, Hila Chalutz-BenGal. ACL ARR August 2026 Submission.

[C5] *Reading Order Inference for Complex Document Layouts*. Iddo Hakim, **Sharva Gogawale**, Omer Ventura, Gal Grudka, Daria Vasyutinsky-Shapira, Berat Kurar-Barakat & Nachum Dershowitz. *ICDAR-DAS 2026* (Best Paper Runner-up) [Paper]

[C4] *A Multi-Stage Agentic Framework for Effective Counter-Narrative Generation and Refinement*. Carmel Kronfeld, **Sharva Gogawale**, Tetsuro Kobayashi, Irad Ben-Gal. *19th International Conference on Natural Language Generation (INLG 2026)*. [Paper]

[C3] *Bag of Bags: Adaptive Visual Vocabularies for Genizah Join Image Retrieval.* **Sharva Gogawale**, Gal Grudka, Daria Vasyutinsky-Shapira, Omer Ventura, Berat Kurar-Barakat, Nachum Dershowitz. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops 2026.* [Paper]

[C2] *Complex Layout Classification in the Wild: A Low-Resource Approach with Layout-Preserving Augmentations.* **Sharva Gogawale**, Iddo Hakim, Gal Grudka, Mohammad Suliman, Omer Ventura, Daria Vasyutinsky-Shapira, Berat Kurar-Barakat, and Nachum Dershowitz. *International Conference on Document Analysis and Recognition (ICDAR 2026), Vienna, Austria.*

[C1] Sharva Gogawale, Madhura Deshpande, Parteek Kumar, Irad Ben-Gal. *Real-Time Attentiveness Detection and Adaptive Feedback in Online Learning Using Facial Expression Recognition and LLMs*. 17th International Conference on Knowledge Discovery and Information Retrieval (Accepted to KDIR 2025). [Paper]

[J1] Sharva Gogawale, Madhura Deshpande, Parteek Kumar, Irad Ben-Gal. *Learner Attentiveness and Engagement Analysis in Online Education Using Computer Vision*. (Accepted to Multimedia Tools and Applications 2026). [Paper]

EXPERIENCE

University of Houston — **Remote**

Graduate Researcher | *Advised by Prof. Rakesh Verma* — *April 2026 - Present*

- Introduced FoSDD, a framework separating non-literal language understanding from deception reasoning through evidence-grounded figure-of-speech representations, in-context learning, and a retrieval-augmented knowledge base with multi-LLM verification **[C7].**
- Developing follow-up methods using input-conditioned **MoE** and **positive-unlabeled learning [C8]** to study whether figurative-language supervision improves cross-domain deception detection.

Tel Aviv University — **Remote**

Research Collaborator | *Advised by Dr. Chen Shani* — *May 2026 - Present*

- Studying concept-aware language model training through next-concept prediction, where probability is assigned across valid concepts rather than only the observed next token, and evaluating how it affects concept coverage, retention, and language modeling performance.
- Investigating how language models organize concepts and encode conceptual representations, using activation patching and circuit discovery to identify the responsible components and trace how conceptual structure emerges across the model.

Tel Aviv University (in collaboration with Waseda University & XPOZ.AI) — **Remote**

Graduate Researcher | *Advised by Prof. Irad Ben-Gal and Prof. Tetsuro Kobayashi* — *April 2025 - March 2026*

- Designed a hybrid human-AI agent framework for automated detection, classification, and targeted mitigation of coordinated misinformation campaigns in online social networks **[C5]**.
- Implemented and refined LLM-based counter-narrative generation agents, optimizing them via empirically validated rhetorical strategies, and evaluating their effectiveness through adversarial multi-agent simulations, demonstrating that refined CNs outperformed a vanilla LLM baseline by an average of 29%.

European Research Council (ERC) Synergy MiDRASH **Tel Aviv, Israel**
AI Researcher | *Advised by Prof. Nachum Dershowitz* *September 2023 - Present*

- Contributing to the €10M ERC Synergy MiDRASH project, developing machine learning methods for medieval manuscript analysis, including handwriting recognition, page segmentation, and intertextual retrieval across large historical collections.
- Exploring **non-literal and conceptual language** in collaboration with Dr. Kfir Bar, spanning multilingual idiom identification, cross-lingual transfer, and connecting abstract morals and ideas to indirect narrative realizations.
- Investigating gender bias in English-to-Hebrew translation, separating grammatical gender constraints from social bias and analyzing whether LLM gender assignments reflect linguistic structure or stereotype-driven patterns in model representations.

LAMBDA: Laboratory for AI, ML, Business & Data Analytics, TAU **Tel Aviv, Israel**
Research Assistant | *Advised by Prof. Irad Ben-Gal & Prof. Parteek Bhatia* *October 2022 - September 2023*

- Developed deep-learning models for learner attentiveness and affective-state analysis, using class-imbalance-aware training and a novel attentiveness formulation, achieving 80.32% engagement accuracy and outperforming prior baselines **[J1]**.
- Integrated LLMs with learner-engagement analytics and lecture-context signals to interpret affective states and generate personalized recommendations for students and instructors, extending the system from passive prediction toward adaptive learning support **[C2].**

TAD: Center for Data Science & Artificial Intelligence **Tel Aviv, Israel**
Graduate Research Assistant *October 2022 - March 2023*

- Developed a BERT-based judicial authorship identification pipeline over U.S. Supreme Court opinions, achieving 76% accuracy and helping secure the $30K TAD 2023 Research Grant.

Google - TAU AI for Social Good **Tel Aviv, Israel**
ML Researcher | *Mentored by Prof. Nachum Dershowitz* *August 2021 - October 2022*

- Developed segmentation, matching, and registration methods for historical Dead Sea Scroll imagery, with outputs integrated into the IAA's public archive.

Undergraduate Researcher, Tel Aviv University
Advanced Segmentation and Word Spotting for Historical Document Recognition
Supervisors: Prof. Lior Wolf & Prof. Nachum Dershowitz

- Developed a historical Hebrew document segmentation and word-spotting pipeline combining top-line estimation, repolygonization, and fully convolutional segmentation, achieving 94.7% F1 at IoU 0.9. (Undergraduate Thesis)

Onshape (PTC Inc.) **Pune, India**
Software Engineering Intern *August 2020 - October 2020*

- Designed, implemented, and deployed scalable backend features for CAD software; delivered a new client-specific module and built ETL workflows with Apache Airflow to support data processing.

TEACHING & MENTORSHIP EXPERIENCE

Head Teaching Assistant, '*Summer Course on Explainable-AI'*: Led instruction and mentorship for 300+ participants from underrepresented STEM communities, delivering a rigorous XAI curriculum in a collaboration between LAMBDA and Women in AI (W-AI), Israel.

AI Research Mentor, '*Postgraduate Institute of Medical Education and Research (PGIMER) & Thapar Institute'*: Mentoring a team developing AI-powered, non-invasive cardiac assessment tools for underdeveloped regions in India; guiding the engineering of text-conditioned SAM segmentation, diffusion-based 3D reconstruction, and LLM-driven diagnostic pipelines for full-text radiology report generation.

ACHIEVEMENTS & HONORS

- **First Place**: Spring 2022 International Startup Hackathon, organized by Coller School of Management & Rutgers University – Initiated an EdTech Startup.
- **Semifinalist:** Annual Startup Competition 2022, TAU - Recognized as Top 10 Tech Track Startups in Israel out of 116 participating ventures.
- **Full M.Sc. Merit Scholarship**: Awarded by the School of Electrical Engineering at Tel Aviv University.
- **Full B.Sc. Scholarship** - Awarded by Tel Aviv University for all four years of undergraduate studies.

SKILLS

NLP & ML: PyTorch, TensorFlow, CUDA, OpenCV, HuggingFace Transformers, PEFT/LoRA, LLM Evaluation, RAG
Programming, ML & Infrastructure: Python, C, C++, Java, R, MATLAB, Git, Docker, Linux, AWS, Apache Airflow